services to stream sensor data to the cloud and provide APIs of their own to allow physical actions to be taken, such as adjusting machinery in a water treatment plant or turning off the lights in your home or office. In this chapter you'll see how to secure the communications of IoT devices when talking to each other and to APIs in the cloud. In chapter 13, we'll discuss how to secure APIs provided by devices themselves.

> **DEFINITION** The ***internet of things*** is the trend for devices to be connected to the internet to allow easier management and communication. ***Consumer IoT*** refers to personal devices in the home being connected to the internet, such as a refrigerator that automatically orders more beer when you run low. IoT techniques are also applied in industry under the name ***industrial IoT*** (IIoT).

12.1 Transport layer security

In a traditional API environment, securing the communications between a client and a server is almost always based on TLS. The TLS connection between the two parties is likely to be end-to-end (or near enough) and using strong authentication and encryption algorithms. For example, a client making a request to a REST API can make an HTTPS connection directly to that API and then largely assume that the connection is secure. Even when the connection passes through one or more proxies, these typically just set up the connection and then copy encrypted